

SQL DATE FUNCTIONS

EXERCISE 5

NOMTHI MAHLANGU

Question: 1

Return each order with the day name of the order date

```
SELECT order-id, customer-id, order-date,  
       DAYNAME(order-date) AS day-name  
FROM orders;
```

Question: 2

Return each customer with the name of the month in which they signed up.

```
SELECT customer-id, customer-name, signup-date,  
       MONTHNAME(signup-date) AS signup-month-name  
FROM customer-signups;
```

Question: 3

Return each sale with the month number extracted from sale_date

```
SELECT sale-id, product-name, sale-date,  
       MONTH(sale-date) AS sale-month
```

```
FROM sales;
```

Question: 4

Return each transaction with the year extracted from transaction_date

```
SELECT transaction-id, customer-id, transaction-date,  
       YEAR(transaction-date) AS transaction-year
```

```
FROM transactions;
```


Question: 5

Return each delivery with the day of the month extracted from delivery_date

SELECT delivery_id, customer_id, delivery_date,

DAY(delivery_date) AS day-of-month

FROM deliveries;

Question: 6

Return all employees and add a column showing today's date.

SELECT employee_id, employee_name, department,

CURRENT_DATE() AS today_date

FROM employees;

Question: 7

Convert the order_date_text string column into a proper date column.

SELECT order_id, customer_id, order_date_text,

TO_DATE(order_date_text) AS order_date

FROM online_orders;

Question: 8

Format payment_date as a text string in the format YYYY-MM-DD

SELECT payment_id, customer_id, payment_date,

TO_CHAR(payment_date, 'YYYY-MM-DD') AS formatted_payment_date

FROM payment_dates;

Question: 9

Calculate how many days have passed since each customer's last purchase. Use today's date.

SELECT customer_id, customer_name, last_purchase_date,

DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase

FROM customer_purchases;

Question: 10

Calculate the expected delivery date by adding 7 days to the order_date.

```
SELECT order_id, customer_id, order_date,  
       DATEADD(DAY, 7, order_date) AS expected_delivery_date  
FROM shipping_orders;
```

Question: 11

Extract the year, month number, and day from booking date into three separate columns.

```
SELECT booking_id, customer_id, booking_date,  
       YEAR(booking_date) AS booking_year,  
       MONTH(booking_date) AS booking_month,  
       DAY(booking_date) AS booking_day  
FROM bookings;
```

Question: 12

Return only orders from the year 2026.

```
SELECT order_id, customer_id, order_date,  
       YEAR(order_date) AS order_year,  
       amount
```

```
FROM yearly_orders
```

```
WHERE YEAR(order_date) = 2026;
```

Question: 13

Return only orders placed in March (month number 3).

```
SELECT order_id, customer_id, order_date,  
       MONTH(order_date) AS order_month,  
       amount
```

```
FROM monthly_orders
```

```
WHERE MONTH(order_date) = 3;
```


Question: 14

Return the last day of the month for each subscription start date

```
SELECT subscription_id, customer_id, start_date,  
       LAST_DAY(start_date) AS month_end_date
```

```
FROM subscriptions;
```

Question: 15

Return the first day of the month for each campaign send date

```
SELECT send_id, customer_id, send_date,  
       DATE_TRUNC('month', send_date) AS month_start_date
```

```
FROM campaign_sends;
```

Question: 16

Format invoice date as a text string showing month name and year

```
SELECT invoice_id, customer_id, invoice_date,  
       TO_CHAR(invoice_date, 'MMMM yyyy') AS invoice_month_year
```

```
FROM invoice_dates;
```

Question: 17

Calculate each customer's age in years using their date of birth and today's date.

```
SELECT customer_id, customer_name, date_of_birth,  
       FLOOR(DATEDIFF(CURRENT_DATE(), date_of_birth) / 365.25) AS customer_age
```

```
FROM customer_birthdays;
```

Question: 18

Classify each order as Weekend/weekday based on the day name of the order date.

Question: 18

```
SELECT order_id, customer_id, order_date,  
       DAYNAME(order_date) IN ('Saturday', 'Sunday') THEN 'Weekend'  
       DAYNAME(order_date) AS day_name,  
       CASE  
         WHEN DAYNAME(order_date) IN ('Saturday', 'Sunday') THEN 'Weekend'  
         ELSE 'Weekday'  
       END AS day_type  
FROM weekend_orders;
```

Question: 19

Extract the quarter number from each transaction date.

```
SELECT transaction_id, customer_id, transaction_date,  
       QUARTER(transaction_date) AS transaction_quarter,  
       amount  
FROM quarterly_transactions;
```

Question: 20

Return only orders that are more than 30 days old from today. Calculate days-since-order using today's date

```
SELECT order_id, customer_id, order_date,  
       DATEDIFF(CURRENT_DATE(), order_date) AS date_since_order,  
       amount  
FROM recent_orders  
WHERE DATEDIFF(CURRENT_DATE(), order_date) > 30;
```

Question: 21

Calculate days since last purchase for each customer using CURRENT_DATE
then use CASE to classify: ≤ 30 days = Active Customer, 31-90 days =
AT Risk Customer, > 90 days = Inactive Customer.

Question 21

SELECT customer_id, customer_name, last_purchase_date,
DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase,

CASE

WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) <= 30 THEN
 'Active Customer'

WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) <= 90 THEN
 'At Risk Customer'

ELSE 'Inactive Customer'

END AS customer_status

FROM customer_recency;

END